

Leakage-Controlled Radiomics Pipeline

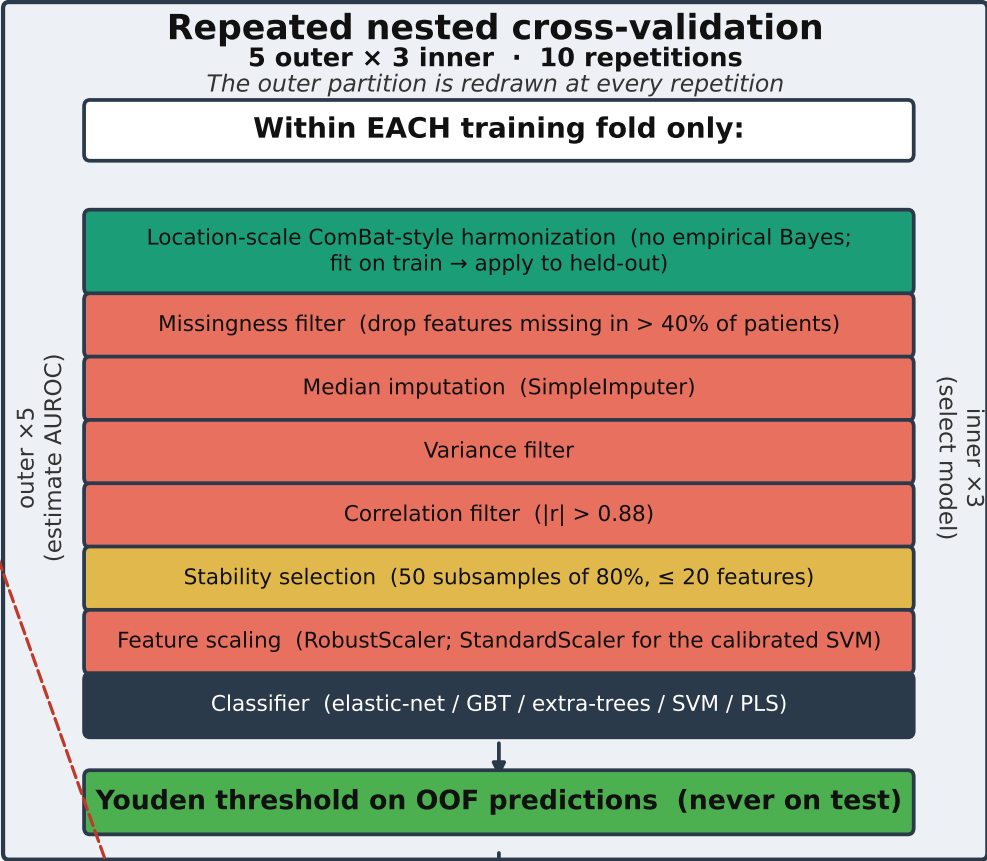
Every label-dependent step is fit inside the training fold only

Public BCBM dataset — 165 patients
T1 post-contrast MRI · expert-reviewed masks · 107 PyRadiomics features

Mask/label canonicalization → lesion-only table (1,841 lesions)
Patient-level aggregation → 642 radiomic · 79 lesion-burden/study · 21 acquisition

Stratified PATIENT-LEVEL split
Train 102 · Tuning 20 · Test 17

Leakage guard:
no patient's lesions
span two splits



LOCKED TEST
(17 patients)
untouched until
final evaluation

Refit on Train + Tuning → evaluate LOCKED TEST once

Deep models (parallel benchmark)
Attention-MIL · 2.5-D CNN · Hybrid fusion
(checkpoint chosen on the tuning set, not test)

External PROXY cohort (OpenBTAI)
applied ONCE; no transform refit on it
(subtype-derived surrogate labels)

Statistical evaluation
Primary: mean AUROC ± SD over 10 partitions · cross-fitted AUROC, patient bootstrap CI (2000×)
Secondary: locked test (n = 16-17) · label permutation · Benjamini-Hochberg FDR per target